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2019, ICRTAC 2019**LiFi for Vehicle to Vehicle Communication – A Review**

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Light Fidelity, also known as LiFi, is a technology based on communication using light as a medium. This technology is known as Visible Light Communication (VLC) which removes the complexity of cable communication. LiFi has evolved over the past years and has been proven to be secure, efficient and can send data at very high rates. The proposed paper reviews the overall foundation of LiFi and studies the possibility of implementing LiFi in a communication system by explaining the architecture, the modulation techniques and its pros and cons. This paper also demonstrates a communication system in which data is sent from a transmitter to a receiver using light as a medium to control the speeds of two motors.

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Keywords: Light Fidelity (LiFi); Visible Light Communication(VLC), wireless communication technologies.

I. Introduction

LiFi or Light Fidelity is a relatively recent technology that refers to a Visible Light Communication (VLC) system which uses light as a pathway for transfer of data at high speeds. Since light is used, it offers a wide range of frequencies and wavelength [1]. This technology can be compared with that of Wi-Fi but offers an increased accessible range, efficiency, security, and much higher speed. LiFi was introduced by Prof. Harald Haas on July 2011 at TED Global Talk.

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In VLC if the LED is glowing, digital data 1 is being transmitted; and if the LED is not glowing, digital data 0 is being transmitted, or no data is being transferred [2, 3]. The LEDs can be switched on and off very rapidly, which can be used to transmit binary data. Hence, if signal processing technology is introduced in the LED circuits, data embedded in the LED beams could be transmitted at very high speeds to the receivers. A series of 1s and 0s carry information that can be easily decoded.

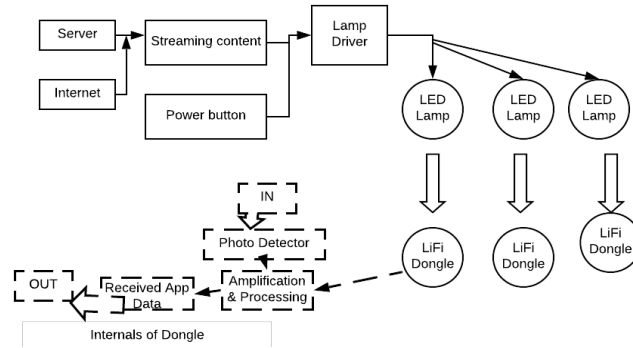


Fig. 1. The layout of LiFi

The transmitter signal will contain flickering of light (that the naked eye will be unable to notice) that can be converted into an electric signal by the receiver. The received signal can then be converted back into its equivalent binary data [4]. This paper reviews state of the art in LiFi technology and presents the work carried out in transmitting data using LiFi by serial communication from an LED transmitter to a Photodiode receiver as shown in Fig. 1.

II. Comparison of wireless communication technologies

In this section, a comparison between the wireless protocols of LiFi, WiFi, Bluetooth and Zigbee are discussed in Table 1[5- 8].

Table 1. Comparison of wireless communication technologies

Feature	WiFi [5 – 8]	LiFi	Bluetooth [5]	ZigBee [5]
Mode of Operation	Using radio waves	Using light waves	Using short wavelength UHF radio waves	Using radio waves
Coverage distance	32m	10m	10m, 100m, Based on classes	10 -100m
Frequency of operation	2.4GHz, 4.9GHz and 5GHz	10000 times radio waves	2.4 – 2.485 GHz	2.4GHz
Speed of transmission	150Mbps	1 Gbps	25Mbps	250 kbit/s

III. Architecture of LiFi

The LiFi architecture is categorized based on layers as shown in Fig. 2 [9]. LiFi consists of three layers i.e. Application layer, MAC layer and Physical layer. The IEEE 802.15.7 protocol defines only two layers, the Physical layer and the MAC layer.

A. IEEE 802.15.7 Protocol

In this IEEE standard, a PHY (Physical 1, 2 and 3) and a MAC (Medium Access Controller) layer for short-range visible light communication in an optically transparent media are defined. The wavelength of visible light spectrum extends from 380nm up to 780nm. This is capable of delivering data high enough to support audio and video multimedia operations and also considers the mobility, compatibility with the infrastructure and noise interference from ambient light.

B. LiFi Physical Layer

The Physical layer or PHY layer is responsible for many things. It handles the transmission and reception of data. There are three layers in the LiFi physical layer [10] as detailed in Table 2.

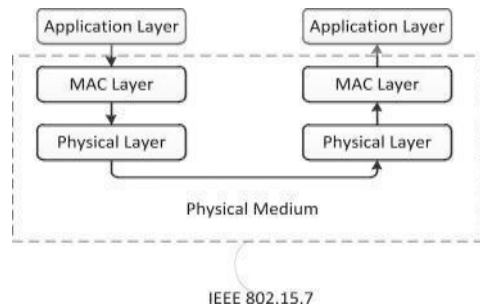


Fig.2. Layered architecture VLC (IEEE 802.15.7) [9]

Table 2. Physical Layers of LiFi

S.No	Operating Mode	Application	Data rate
1	PHY I layer	Outdoor application	11.67 to 267.6 kbit/s
2	PHY II layer	Indoor application	1.25 to 96 Mbit/s
3	PHY III layer	Colour Shift Keying (CSK)	12 to 96 Mbit/s

C. LiFi MAC Layer

There are three MAC layers:

1. Peer to peer: Communication between two devices where one of them acts as a coordinator.
2. Star Communication: Communication between several devices. Here, one of them acts as a coordinator and this device is used as an illumination infrastructure.
3. Broadcast One: In this communication, the coordinator sends data to multiple devices. This type of communication is unidirectional.

IV. Modulation Techniques

The modulation signals are used to control the flickering of the Light Emitting Diode at different frequencies that contain the data to be transmitted. Modulation techniques are required to carry out the communication even if the intensity of light or the illumination of light is not available. The intensity of light of the LED and the frequency of the flickering of the LED depends on the information carried in the message signal. The three types of modulation [11] in LiFi are: Single Carrier Modulation (SCM), Multiple Carrier Modulation (MCM) and Colour Modulation or Colour Shift Keying (CSK).

A. Single Carrier Modulation (SCM)

SCM is more suitable for application with low to moderate data rates. SCM is a modulation technique in which any number of signals can be received but they are all modulated individually at different frequencies [12]. The Modulation techniques in SCM are ON-OFF Keying (OOK), Pulse-amplitude modulation (PAM) and Pulse position modulation (PPM).

- **OOK (ON-OFF Keying):** In ON-OFF Keying, data is represented by on and off of the LED. LED ON means logic '1' and LED OFF means logic '0'. So, data is transmitted in the form of 1s and 0s through the flickering of the LED. This method of modulation is adopted in the proposed project where data transmission is done using LEDs.
- **Pulse-amplitude modulation (PAM):** In this form of modulation, the message carrying the information is encoded in the amplitude of a continuous series of signal pulses.
- **Pulse Position Modulation:** This modulation provide an efficient and dimming support in variable pulse position modulation (**VPPM**)

B. Multiple Carrier Modulation (MCM)

The disadvantage of SCM is that it requires complex processes to handle information at higher data rates. For higher data rates, MCM was developed. There are two types of MCMs i.e. Orthogonal Frequency Division Modulation (OFDM) and Colour Shift Modulation or Colour Shift Keying (CSK).

1. OFDM

This modulation is mainly used when more than one devices used as transmitters. This reduces shadowing effect. This effect can be removed by changing the bandwidth of the system to a greater frequency.[13 – 17]

2. Colour Shift Modulation

The Colour Shift modulation scheme is used to transmit different data using different colours of light having different wavelengths. LEDs of Red, Blue and Green are used to send data carrying different information. The receiver is programmed to identify the different wavelengths of the messages and decode them.

V. Advantages of LiFi

The advantages of LiFi are mentioned below:

1. Speed: LiFi can transmit data at the rate of 10 Gbit/s. This is almost 250 times faster than any other high speed broadband connection.
2. High-density coverage: LiFi is more ideal for a high density coverage. WiFi is more ideal for common purposes.
3. Secure: LiFi is more assured and dependable than WiFi. Light cannot pass through opaque objects hence it can be easily blocked by many surrounding objects.
4. Cost: LiFi technology is free.
5. Larger Spectrum: There are no restrictions to the capacity of LiFi. Visible light spectrum is about 10,000 times larger than RF spectrum.

VI. Challenges of LiFi

Although LiFi has many advantages, it poses some limitations as well.

1. The modulation in LiFi is a challenge when the light illumination is low.
2. LiFi cannot provide an Internet connection when there is no light source. This may cause limitations for locations and situations where LiFi may be necessary.
3. Light cannot pass through physical barriers. This may cause certain limitations.
4. The major drawback of LiFi is it can be used only within the small area as it cannot penetrate through walls.

VII. Applications of LiFi

The LiFi system is still in its experimental phase towards the future of self automated cars. These cars can operate using communication between vehicles and street lamps. Also access to high speed internet using street lamps can be implemented. Once an efficient and secure design for visible light communication is ready, it can change the future of communication. LiFi can also be used in disaster management and Medical environments where the use of RF waves may pose as a risk for patients.

A. Vehicle to Vehicle communication (V2V Communication)

LiFi can be used in communication between vehicles using their headlights and this can be used to reduce accidents occurring on the roads. A demo model was implemented to demonstrate the VLC in a stationary scenario. The objective of the work was to control the speed of a DC motor, which is connected to the Receiver (as shown in Fig. 4), by using the speed of another DC motor which is connected to the Transmitter (as shown in Fig. 3). Arduino Uno microcontroller was used to connect all peripherals, it is easy to program and is low cost [18, 19].

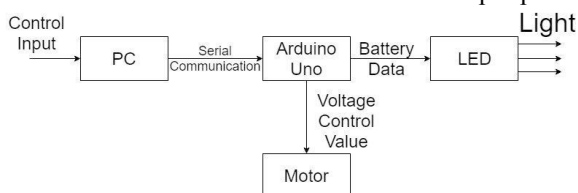


Fig. 3. Block diagram of Transmitter

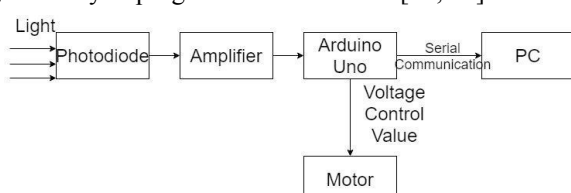


Fig.4. Block diagram of Receiver

The Fig. 5 shows the complete schematic diagram of the project designed to create a communication system. Fig. 6 shows the complete hardware setup. Speed of the motors is controlled by Arduino using its analog pins by varying the voltage across the terminals in the range of (0-5) V. The speed of motor 1 is varied and the speed data is transmitted by toggling the red LED connected to Arduino 1. This light is received by a photodiode which is connected to Arduino 2 and the data is retrieved and the motor 2 speed is controlled. The algorithms of the transmitter and receiver controller are given in Fig. 10. The interval is defined as 25000 which is in terms of microseconds. Timer0 of the Arduino is used to control the motor speed, the Arduino UART is connected to a PC using serial communication, and the control signals are sent to the Arduino from the PC, and the control value is sent by the Arduino and is displayed in the terminal on the PC.

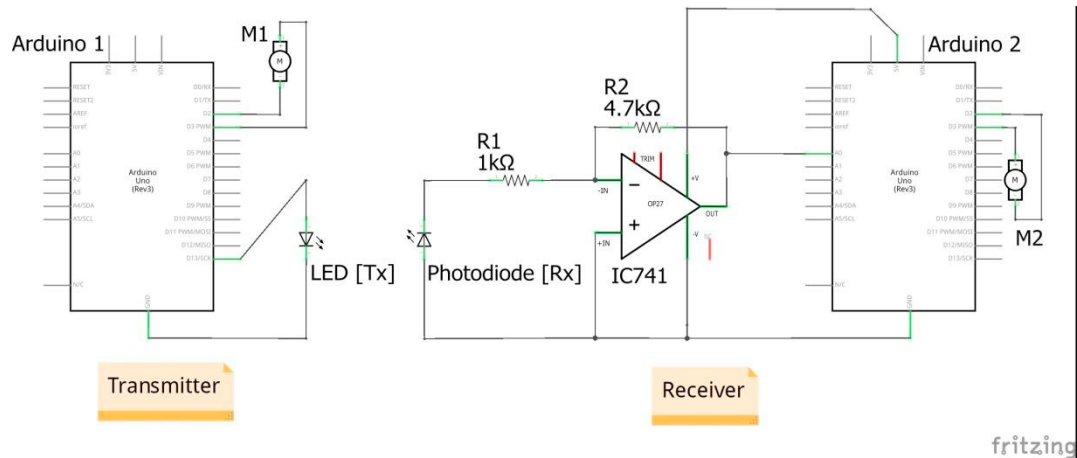


Fig.5. Schematic diagram.



Fig.6. Hardware setup.

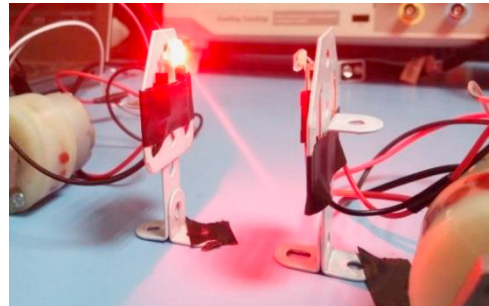


Fig.7. Communication between the LEDs.

The real time communication is shown in Fig 7. In vehicles, such a communication can help cars communicate their speeds and other parameters and hence prevent accidents and decrease traffic. To study and analyze the working of the receiver side of the system, it was implemented as a Simulink model on MATLAB. The model is shown on Fig 8. The PWM signal, representing the data, is fed to the photodiode as light signal with a threshold value, as shown in figure 9(a). There is a downward shift in the signal which is due to the threshold value. This output voltage is amplified by the inverting amplifier and output response is observed in figure 9(b). The upward shift in the waveform is due to the threshold value which is 0.25V (-ve due to inverting amplifier). The algorithm of the receiver controller is given in the Fig. 10. Here, the same interval value is set to synchronize it with the transmitter. Timer 0 is initiated and serial communication is set with the PC on the receiver end. The received value is converted to decimal; this value is used to control the motor in the receiving end. Threshold value is the voltage detected at the photodiode output that is due to the presence of lighting in the environment of where the communication is happening [8]. After this value is determined, it is considered as a digital LOW for the data transmission and anything fairly above this value is considered as a digital HIGH. The bit stream received is stored in an integer array and then converted to decimal value. The communication thus received is as shown in Fig.7.

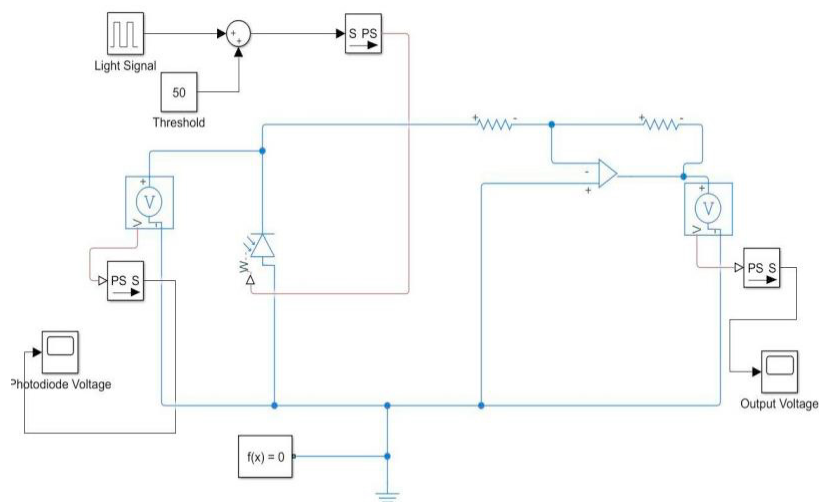


Fig.8. Simulation of receiver side.

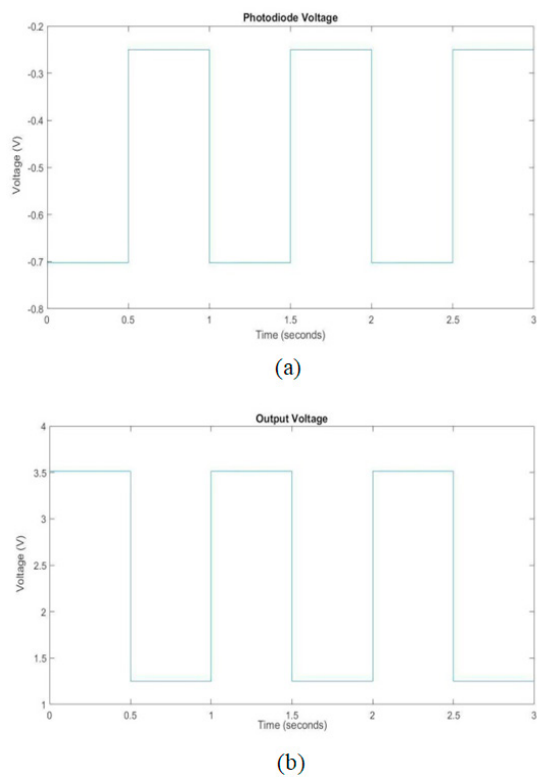


Fig. 9. (a).Photodiode output waveform (b).Output waveform of the inverting amplifier

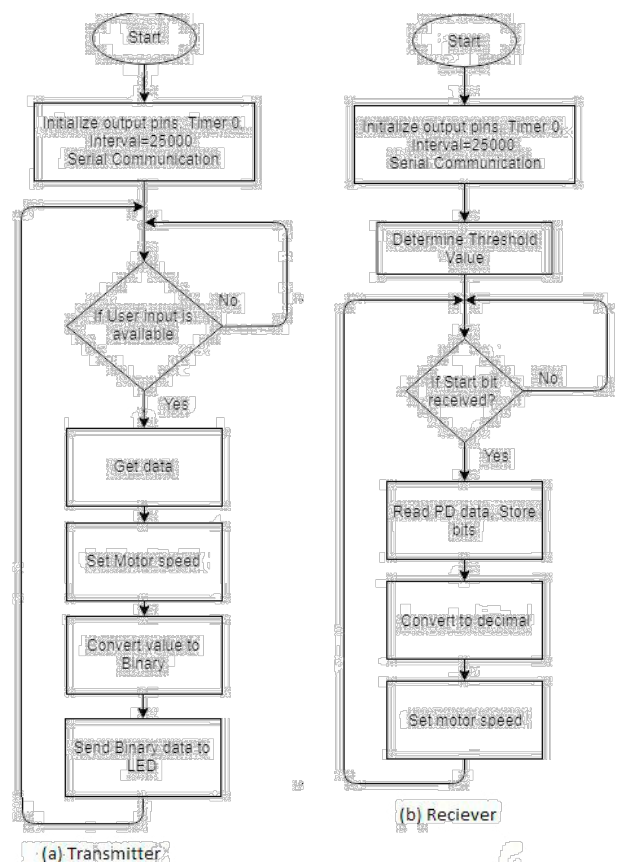


Fig. 10. Transmitter & Receiver signal flow.

VIII. Conclusion

In this paper, LiFi is introduced as a communication system with its modulation techniques and complete architecture explained. The challenges and advantages of LiFi are outlined with its purpose to provide high speed data transmission being one of its biggest pros. The implementation of LiFi is introduced as an application V2V Communication the project performed in which the speed of two motors is controlled using LiFi. Data is transmitted from one control unit to the other and this controls the speed of the motor. The necessary analysis of the system with the block diagrams is also shown. It is hence concluded that LiFi is used as a communication tool and the control of two motor speeds is performed.

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